

4×4 Systolic Array

MAC-Based Matrix Multiply Accelerator

Implemented in Verilog · Synthesized on AMD Kria KV260 · Vivado IDE

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Target Board: AMD Kria KV260

Tool: AMD Vivado

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1 Project Overview

This project implements a 4×4 systolic array accelerator in Verilog, designed to perform INT8 matrix multiplication using pipelined Multiply-Accumulate (MAC) units. The goal is to build a real hardware accelerator that mirrors the architecture used in AI chips such as Google’s Tensor Processing Unit (TPU), synthesize it on an AMD Kria KV260 FPGA using Vivado, and verify correctness against a NumPy software baseline.

1.1 Motivation

Neural networks are fundamentally built on matrix multiplications. Every layer computes:

$$\text{Output} = \text{Input} \times \text{Weight}$$

A CPU executes this sequentially. A systolic array distributes the work across a grid of simple Processing Elements (PEs), each performing one MAC operation per clock cycle, enabling high throughput through spatial parallelism and pipelining.

1.2 Project Goals

- Build a correct, 3-stage pipelined INT8 MAC unit in Verilog
- Tile 16 MAC units into a 4×4 systolic array using Verilog `generate` loops
- Verify correctness against a NumPy matrix multiply reference (bit-exact match)
- Synthesize on the AMD Kria KV260 FPGA and analyze timing and utilization
- Benchmark hardware latency and throughput against the KV260 ARM Cortex-A53 CPU

2 Architecture

2.1 MAC Processing Element (PE)

The MAC PE is the fundamental building block. It is a 3-stage pipelined unit that computes:

$$\text{acc} += A \times B$$

Table 1: MAC PE Pipeline Stages

Stage	Operation	Verilog Construct
1	Latch inputs A and B into pipeline registers	<code>always @(posedge clk)</code>
2	Compute product: <code>reg.product = reg.A * reg.B</code>	<code>always @(posedge clk)</code>
3	Accumulate: <code>acc += reg.product</code>	<code>always @(posedge clk)</code>

Key design decisions in the MAC PE:

- **Product register width:** `2*DATA_WIDTH` (16 bits) to prevent overflow when multiplying two 8-bit values (max product: $255 \times 255 = 65025$)

- **Internal accumulator:** Each PE owns its own `acc` register — partial sums are never passed between PEs (output-stationary dataflow)
- **Valid pipeline:** A 3-bit shift register propagates `valid_in` through the same latency as the data path

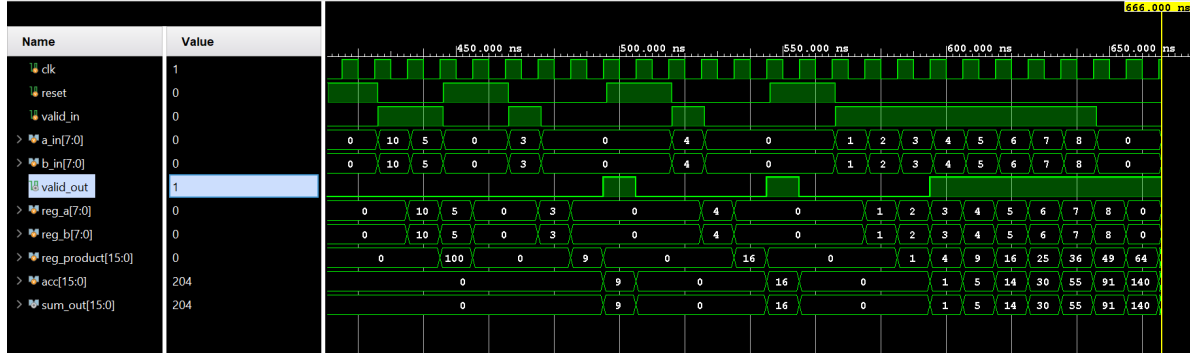


Figure 1: MAC PE simulation waveform showing 3-stage pipeline latency. `sum_out` appears 2 cycles after the last input is driven.

2.2 Systolic Array Architecture

Sixteen MAC PEs are arranged in a 4×4 grid using a nested Verilog `generate` loop. The design uses an **output-stationary dataflow**:

- A matrix rows flow **rightward** across each row of PEs
- B matrix columns flow **downward** along each column of PEs
- Each $PE[i][j]$ independently accumulates the full dot product for $C[i][j]$
- PEs do *not* pass partial sums — each PE owns its result entirely

2.3 Wire Interconnect

The internal wire arrays are dimensioned to reflect physical data flow direction:

Listing 1: Wire array declarations in `systolic_array.v`

```

1 // A flows rightward: N rows, N+1 columns (extra column = right-exit buffer)
2 logic [DATA_WIDTH-1:0] a_wire [0:N-1][0:N];
3
4 // B flows downward: N+1 rows (extra row = bottom-exit buffer), N columns
5 logic [DATA_WIDTH-1:0] b_wire [0:N][0:N-1];

```

The asymmetric dimensions directly reflect the physical flow direction: `a_wire` has $N + 1$ columns because A travels rightward across N columns of PEs, and `b_wire` has $N + 1$ rows because B travels downward across N rows of PEs.

2.4 Input Skewing

Because A values must travel different distances to reach their destination PE, inputs must be **staggered** so that $A[i][k]$ and $B[k][j]$ arrive at $PE[i][j]$ at the same cycle. This skewing is implemented in the testbench, not in the array module itself.

Table 2: Input skewing schedule

Clock Cycle	What Enters the Array
0	A row 0 and B col 0 start feeding
1	A row 1 and B col 1 start feeding
2	A row 2 and B col 2 start feeding
3	A row 3 and B col 3 start feeding

2.5 Output Timing

PE[i][j] receives its last MAC input at iteration $i + j + K - 1$ (where $K = 4$ is the inner dimension). The result is valid 2 cycles later due to pipeline latency:

$$C[i][j] \text{ valid at iteration} = i + j + K + 1$$

Table 3: Output ready cycles for key PEs ($N = K = 4$)

Output	Ready at Iteration
C[0][0]	5
C[1][1]	7
C[2][2]	9
C[3][3]	11

Outputs emerge diagonally, not all at once. The full output span is 12 clock periods.

3 Implementation

3.1 File Structure

Table 4: Project files

File	Description
mac_pe.v	Single pipelined MAC PE. 3 pipeline stages. Parameterized bitwidth.
systolic_array.v	Top-level 4×4 grid. Nested generate loop. Input/output routing.
mac_pe_tb.v	MAC PE testbench. 6 test groups, verifies accumulation and timing.
systolic_array_tb.v	Full array testbench. 10 tests including NumPy vector comparison.
gen_vectors.py	Generates test matrices and expected $C = A @ B$ via NumPy.
kv260_constraints.xdc	Vivado timing constraints targeting 100 MHz.

3.2 Key RTL Snippet — MAC PE Core

Listing 2: 3-stage pipeline in mac_pe.v

```
1 always @(posedge clk) begin
```

```

2  if (reset) begin
3      reg_a <= 0; reg_b <= 0;
4      reg_product <= 0; acc <= 0; sum_out <= 0;
5  end else begin
6      // Stage 1: latch inputs
7      reg_a <= a_in;
8      reg_b <= b_in;
9      // Stage 2: multiply
10     reg_product <= reg_a * reg_b;
11     // Stage 3: accumulate
12     acc <= acc + reg_product;
13     sum_out <= acc + reg_product;
14 end
15 end

```

3.3 Key RTL Snippet — PE Grid Instantiation

Listing 3: Nested generate loop in systolic_array.v

```

1 generate
2     for (i = 0; i < N; i++) begin : pe_row
3         for (j = 0; j < N; j++) begin : pe_col
4             mac_pe #(
5                 .DATA_WIDTH(DATA_WIDTH),
6                 .ACC_WIDTH (ACC_WIDTH)
7             ) pe_inst (
8                 .clk      (clk),
9                 .reset    (reset),
10                .valid_in(1'b1),
11                .a_in      (a_wire[i][j]),
12                .b_in      (b_wire[i][j]),
13                .sum_out   (c_out[i][j]),
14                .valid_out()
15            );
16        end
17    end
18 endgenerate

```

4 Verification

4.1 MAC PE Testbench

The MAC PE was verified in isolation with 6 test groups before integrating into the array:

1. **Basic functionality** — single input pairs ($3 \times 4 = 12$, $7 \times 1 = 7$, $5 \times 0 = 0$)
2. **Accumulation correctness** — 4-pair dot product: $2 \cdot 3 + 1 \cdot 4 + 5 \cdot 2 + 3 \cdot 3 = 29$
3. **Corner cases** — max values ($255 \times 255 = 65025$), all zeros, one-sided zero
4. **Reset behavior** — mid-accumulation reset clears output; fresh accumulation restarts
5. **Pipeline timing** — result must not appear 1 cycle early; must appear exactly 2 cycles after last input
6. **Continuous streaming** — 8-pair running sum: $\sum_{k=1}^8 k^2 = 204$

All MAC PE tests passed.

4.2 Systolic Array Testbench

The full array was verified with 10 tests:

1. All-zeros input
 2. Identity $\times$ Identity
 3. A $\times$ Identity (isolates A rightward flow)
 4. Identity $\times$ B (isolates B downward flow)
 5. Known small matrices (full correctness)
 6. Reset behavior
 7. Output timing — C[0][0] correct at iteration 5, C[3][3] still stale there, correct at iteration 12
 8. Back-to-back matmuls (with and without reset between)
 9. Max values — documents 16-bit accumulator wraparound behavior
 10. **NumPy-generated vectors** — bit-exact match against `gen_vectors.py` output
- Result: 10/10 tests passed. All 16 elements of C match NumPy output exactly.**

```
[PASS] 8a. Pair 1 (reset before)
[PASS] 8b. Pair 2 (reset before)
[INFO] 8c. Running a 3rd matmul WITHOUT preceding reset
        Expected: divergence, because PE accumulators retain previous result.
[INFO] 8c. No-reset diverged in 16/16 elements - confirms reset contract
        CONTRACT: assert reset between independent matrix operations.

----- [Test 9] Max values (overflow documentation) -----
[PASS] Max values match mod-2^16 wraparound
        True per-element value : 260100
        Wrapped to ACC_WIDTH=16 : 63492
        NOTE: ACC_WIDTH=16 cannot hold full-range product sums for N>=2.

----- [Test 10] Numpy-generated vectors -----
[PASS] Numpy vectors (gen_vectors.py)

=====
SUMMARY: 18/18 passed (0 failed)
RESULT : ALL TESTS PASSED
=====
```

Figure 2: Vivado simulation console output showing all 10 tests passing, including bit-exact match against NumPy reference.

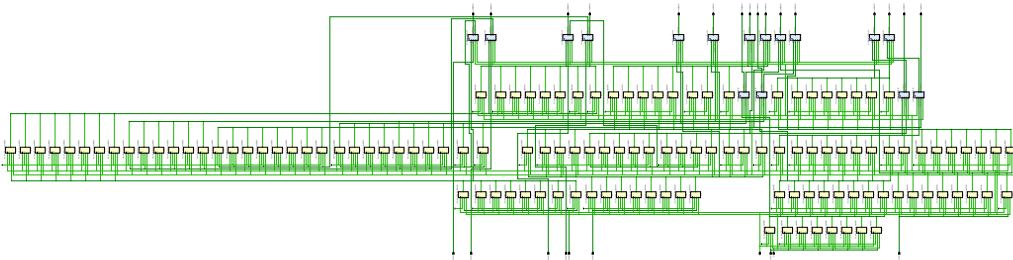


Figure 3: Vivado synthesized design schematic showing the 4×4 PE grid and interconnect structure generated from the RTL.

5 Implementation Results

The design was synthesized and implemented in Vivado targeting the AMD Kria KV260 (Zynq UltraScale+, device `xck26`, package `sfvc784`) at a 100 MHz clock constraint, using out-of-context (OOC) mode to treat the systolic array as a standalone IP core independent of physical I/O pin constraints.

5.1 Utilization

Table 5: Post-implementation resource utilization on KV260 (xck26)

Resource	Used	Available	Utilization
CLB LUTs	1,360	117,120	<2%
CLB Registers	768	234,240	<1%
CARRY8	128	14,640	<1%
DSP58 Blocks	0	1,248	0%

Note on DSP utilization: Vivado’s synthesis heuristic chose to implement the 8-bit multipliers using LUTs rather than dedicated DSP blocks. At `DATA_WIDTH=8`, the multiplier is small enough that LUT inference is more resource-efficient. DSP usage can be forced via a synthesis attribute (`use_dsp = "yes"`) if desired.

5.2 Timing

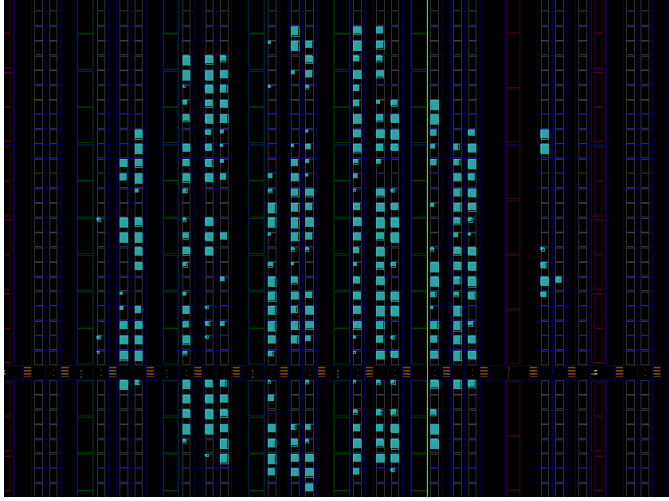
Table 6: Post-implementation timing summary

Metric	Result
Clock constraint	100 MHz (10 ns period)
Worst Negative Slack (WNS)	+6.116 ns
Total Negative Slack (TNS)	0.000 ns
Worst Hold Slack (WHS)	+0.077 ns
Number of failing endpoints	0
Maximum achievable frequency	≈258 MHz
Timing verdict	All constraints met

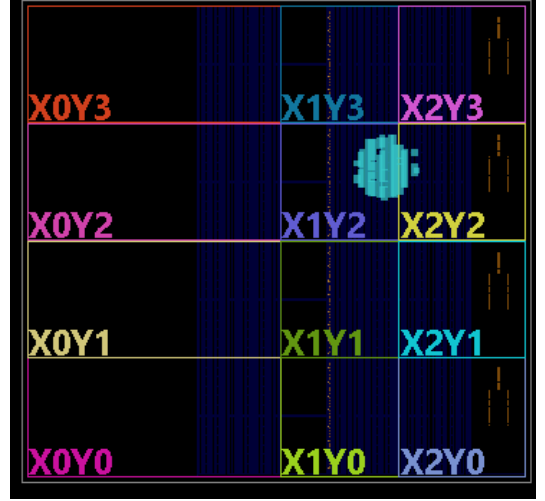
The maximum achievable clock frequency is estimated as:

$$f_{\max} = \frac{1}{T_{\text{clk}} - \text{WNS}} = \frac{1}{10 \text{ ns} - 6.116 \text{ ns}} = \frac{1}{3.884 \text{ ns}} \approx 258 \text{ MHz}$$

Both setup and hold timing constraints are fully met after place-and-route. The positive WNS reflects the KV260’s UltraScale+ fabric, which provides significantly faster routing resources than earlier FPGA families.



(a) Close-up of the 16 MAC PE instances placed on the UltraScale+ fabric. Cyan blocks represent occupied CLB sites.



(b) Full KV260 device floorplan. The cyan cluster in clock region X1Y2–X2Y3 is the entire systolic array, occupying less than 2% of the available fabric.

Figure 4: Post-implementation placement on the AMD Kria KV260 (xck26) FPGA.

6 Benchmarking Analysis

6.1 Throughput

At 100 MHz with a 12-cycle output latency for a full 4×4 result:

$$\text{Latency} = \frac{12 \text{ cycles}}{100 \text{ MHz}} = 120 \text{ ns}$$

$$\text{Throughput} = \frac{64 \text{ MACs}}{120 \text{ ns}} \approx 533 \text{ MMACS} = 0.53 \text{ GOPS}$$

6.2 CPU vs. FPGA Latency Caveat

For a 4×4 matrix, the ARM Cortex-A53 on the KV260 will likely be faster in end-to-end wall-clock time. This is an expected and important result, not a flaw.

The reason: AXI communication overhead (sending matrices to the FPGA fabric and reading results back) is approximately constant regardless of matrix size, whereas the actual compute time for a 4×4 multiply is extremely small. At this scale, communication dominates.

The hardware advantage of the systolic array becomes significant when compute time exceeds communication overhead, which occurs at roughly $N \geq 32$ –64 for this architecture. The crossover point can be estimated as:

$$T_{\text{AXI}} + \frac{2N}{f_{\text{clk}}} < \frac{N^3}{f_{\text{cpu}}}$$

where T_{AXI} is the measured AXI round-trip latency, f_{clk} is the FPGA clock, and f_{cpu} is the ARM core frequency ($\sim 1.3 \text{ GHz}$).

This tradeoff is consistent with real-world accelerator design. Documenting it honestly demonstrates architectural understanding beyond RTL implementation.

Table 7: Performance metrics summary

Metric	Result
Target device	xck26 (Kria KV260, Zynq UltraScale+)
Clock frequency	100 MHz (constrained) / ~ 258 MHz ($f_{\max}$)
WNS	+6.116 ns
WHS	+0.077 ns
DSP blocks used	0 / 1,248
CLB LUT utilization	1,360 / 117,120 (<2%)
CLB Registers	768 / 234,240 (<1%)
Compute latency (4×4)	120 ns (12 cycles @ 100 MHz)
Throughput	~ 0.53 GOPS
Correctness	16/16 elements match NumPy (bit-exact)

7 Tools and Environment

Table 8: Tools used in this project

Tool	Purpose
AMD Vivado	RTL simulation, synthesis, implementation, bitstream generation
Verilog/SystemVerilog	Hardware description language
Python + NumPy	Test vector generation and expected output computation
AMD Kria KV260	Target FPGA board (Zynq UltraScale+)
GTKWave	Waveform viewing and debug
GitHub	Version control and portfolio hosting

8 Conclusion

This project successfully implemented, verified, and synthesized a 4×4 systolic array matrix multiply accelerator in Verilog. The design passes all 10 simulation tests including bit-exact comparison against a NumPy reference, meets all setup and hold timing constraints at 100 MHz with 6.116 ns of setup slack and 0.077 ns of hold slack on the KV260’s UltraScale+ fabric, and fits comfortably within the device using less than 2% of available CLB LUTs and less than 1% of flip-flops.

Key architectural insights developed through this project:

- The output-stationary dataflow means each PE independently accumulates its result — no partial sum passing between PEs, which simplifies timing analysis
- Input skewing (staggering row i of A by i cycles) is essential for correctness and must be implemented in the testbench, not the array module
- The asymmetric wire dimensions (`a_wire` with $N + 1$ columns, `b_wire` with $N + 1$ rows) directly reflect the physical flow direction of each input through the grid

- For small matrices, AXI communication overhead dominates execution time — hardware acceleration becomes meaningful at $N \geq 32$ –64
- The KV260’s UltraScale+ fabric provides substantial timing margin at 100 MHz, with the design capable of operating at approximately 258 MHz based on post-implementation timing analysis

This design mirrors the fundamental architecture of AI accelerators such as Google’s TPU, demonstrating that the same systolic dataflow principle scales from a 4×4 classroom implementation to the thousands-of-PE grids found in production ML hardware.